hsc_chemistry_textbook

Maharashtra State Board Curriculum | Academic Year 2025-2026

Grade Level: 12th Standard | Subject: Chemistry | Resource Category: TEXTBOOK
Storage Path: QpGen_dataset / 12th Science / Chemistry / Textbook

SECTION 1: COURSE OVERVIEW & UNIT OBJECTIVES

1.1 Systematic study notes designed in alignment with latest MSB Board Curriculum guidelines.
1.2 Comprehensive coverage of fundamental concepts, derivations, and board examination patterns.
1.3 Includes step-by-step solved numerical problems, labeled diagrams, and key definitions.

SECTION 2: CORE CONCEPTS & THEORETICAL FOUNDATIONS

Topic 1: Fundamental Physical Principles & Laws
- Detailed derivation of core equations and state variable relations.
- Analysis of boundary conditions, experimental setups, and observational outcomes.
- Conceptual clarity required for MHT-CET, HSC Board, and competitive entrance evaluations.

Topic 2: Analytical Methods & Mathematical Formulations

- Expression 1: Differential and integral forms applicable to curriculum problem sets.
- Expression 2: Standard vector representations and dimensional analysis paramete

PAGE 2: KEY FORMULAS, DEFINITIONS & SOLVED EXAMPLES

2.1 Essential Formulas & Constant Table

[Eq 1.1] Primary Equation: $F = m * a$ | Unit: Newton (N) | Dimension: $[M^1 L^1 T^{-2}]$
[Eq 1.2] Energy Relation: $E = h * f$ | Planck Constant $h = 6.626 \times 10^{-34}$ J.s
[Eq 1.3] Wave Equation: $v = f * \lambda$ | Velocity of Propagation in Medium

2.2 Important Technical Definitions

Definition 1: Principle of Superposition - When two or more waves overlap, the resultant displacement is equal to the vector sum of individual displacements at that point.

Definition 2: Conservation Law - Total scalar energy in an isolated thermodynamic system remains constant over time under all state transformations.

2.3 Step-by-Step Solved Board Problem

Problem Statement: Calculate the total work done when a system expands reversibly from V_1 to V_2.
Solution Step 1: Write down the given initial parameters $V_1 = 2.0$ L, $V_2 = 5.0$ L, $P = 1.013 \times 10^5$ Pa.
Solution Step 2: Apply $W = - P_{ext} * (V_2 - V_1) = - 1.013 \times 10^5 * (3.0 \times 10^{-3}) = -303.9$ Joules.Final Result: Total Work Done by

PAGE 3: PREVIOUS YEAR BOARD QUESTIONS & PRACTICE SET

3.1 Board Examination Questions (2020 - 2025)

Q1 [2 Marks]: State and explain Huygens' principle of wave propagation with a clean diagram.
Q2 [3 Marks]: Derive an expression for the work done in rotating a magnetic dipole in a uniform magnetic field.
Q3 [4 Marks]: Explain the working mechanism of a PN Junction Diode in forward and reverse bias modes.

3.2 Self-Assessment Practice Problems for Students

1. A coil of inductance 0.5 H and resistance 100 ohms is connected to a 240 V, 50 Hz AC supply. Find peak current.
2. Calculate the de Broglie wavelength associated with an electron accelerated through a potential difference of 100 V.
3. Solve the differential equation dy/dx + y*tan(x) = sec(x) with initial condition y(0) = 1.